

Optics Letters

Time transfer through the optical supervisory channel in wavelength division multiplexing systems

HANHUANG CHENG, GUILING WU,*  FAXING ZUO,  LIANG HU,  AND JIANPING CHEN

State Key Laboratory of Advanced Optical Communication Systems and Networks, Department of Electronic Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

*Corresponding author: wuguilin@sjtu.edu.cn

Received 6 August 2019; revised 19 September 2019; accepted 22 September 2019; posted 30 September 2019 (Doc. ID 374540); published 18 October 2019

We present an optical fiber time transfer scheme through the optical supervisory channel (OSC) in wavelength division multiplexing (WDM) systems. A sub-band of the standard OSC band is used to transmit time signals by only adding sub-OSC filters in commercial WDM systems. The transmission of the existing communication data service is not affected. The wavelengths of bidirectional time transmission in the same sub-band can be very close, or even the same, to maximize the bidirectional propagation symmetry and improve the precision of time transfer. Combined time, optical supervisory signal, and communication data transmission is experimentally demonstrated along a single 100 km optical fiber in a WDM system in laboratory. © 2019 Optical Society of America

<https://doi.org/10.1364/OL.44.005206>

Optical fiber transmission has the advantages of low loss, large capacity, high reliability, etc., and it has been widely used in the field of communication [1]. The use of dedicated fiber-optic links enables high-precision time transfer, but the cost of laying or leasing dedicated fiber-optic links is high, severely limiting the use of large-scale, high-precision time transfer. The optical fiber time transfer through the widely distributed optical fiber communication networks can greatly save the cost and is an ideal choice for realizing high-precision and large-scale optical fiber time transmission. Network Time Protocol (NTP) and Precision Time Protocol (PTP) are time synchronization technologies [2,3] that are widely used in existing networks. However, the time synchronization precision of NTP and PTP can only reach the order of milliseconds (ms) and microseconds (μ s), respectively, which cannot meet the requirements of ultra-high-precision time synchronization in future communication fields. The Institute of Electrical and Electronics Engineers (IEEE) 1588v2 protocol can only achieve precision on the order of sub- μ s [4], which is mainly related to the network load and link asymmetry. In 2016, Nuijts *et al.* proposed a network synchronization and frequency distribution scheme [5], which carries bidirectional time frequency signals in the 1470 nm band (1460–1480 nm) and the

1490 nm band (1480–1500 nm), respectively, and transmits them with optical supervisory signal and communication data in a single fiber by coarse wavelength division multiplexing (WDM). Large wavelength differences for the forward and backward time transmission will lead to severe asymmetry of bidirectional transmission delays, and link calibration is indispensable, which is complex especially for long-distance transmission [6,7]. In addition, to perform the proposed time transfer scheme in the existing commercial wavelength division multiplexing networks, the optical supervisory channel (OSC) filters employed in fiber links must be replaced by dedicated coarse wavelength division multiplexers. It will not only lead to a high cost, but also the interruption of the existing communication service. Furthermore, the method occupies two coarse wavelength division bands that can be used for communication services to support the rapid growth of communication services.

In this Letter, we propose an optical fiber time transfer scheme through the OSC band (1500–1520 nm) [8,9] in commercial WDM systems [10]. Different sub-bands in the OSC band are employed to transmit the time signal and optical supervisory signal, respectively. The two signals are multiplexed by a sub-OSC filter and transmitted over an optical fiber with the optical communication data through commercial OSC filters. There are several advantages to transmit time signals using the sub-OSC band. First, the wavelengths for two directions in the same sub-OSC band can be set very close, or even the same, to ensure maximum bidirectional symmetry of the link delay and achieve a high-precision time transfer without requiring link calibrations. Second, the proposed scheme can be installed in the commercial WDM networks without interrupting the existing communication services, since the OSC filters in fiber links are not needed to be replaced. It is very important for network carriers and clients and can dramatically reduce the obstacle to realize high-precision time transfer in the widely covered commercial WDM networks. The proposed method also does not occupy additional wavelength resources for communication data services. Time transfer in a 100 km WDM system with 10 Gbit/s data channels is experimentally demonstrated. A time transfer with a peak-to-peak value of less than 142 ps and time deviations of less than 15 ps/s and 2 ps/ 10^4 s

are reached. The observed bit error rates (BERs) of OSC and the data channel are less than 7.46×10^{-14} and 1.157×10^{-15} , respectively, which meet WDM optical transmission standards [11,12].

The proposed time transfer through OSC in WDM systems is schematically shown in Fig. 1(a). The system consists of three parts: time transfer, optical supervisory signal transmission, and communication data transmission. The communication data are transmitted over the communication band, such as the C band, and are multiplexed/demultiplexed with the OSC band through OSC filters just as commercial WDM systems. The OSC band is separated into two sub-channels by sub-OSC filters for transmitting time signals and optical supervisory signals, respectively, as shown in Fig. 1(b). In the figure, the time signal is carried on the wavelengths in the sub-OSC band ($\lambda_1 \sim \lambda_2$), and the remaining wavelengths in the OSC band are used to transmit the optical supervisory signal. Considering the operation wavelength range of most commercial small form-factor pluggables (SFPs) for existing OSC signal transmission are less than 13 nm in their operating temperature range [13,14], one can find an extra sub-band in the 20 nm OSC band for time transfer, while still keeping the optical supervisory signal transmission as that in the commercial WDM systems. In intermediate nodes, the time transfer signal, communication data, and optical supervisory signal are multiplexed/demultiplexed in the optical domain and processed separately. Therefore, one can see that by only adding sub-OSC filters at each node, time transfer can be performed in existing commercial WDM systems without interrupting the existing communication services.

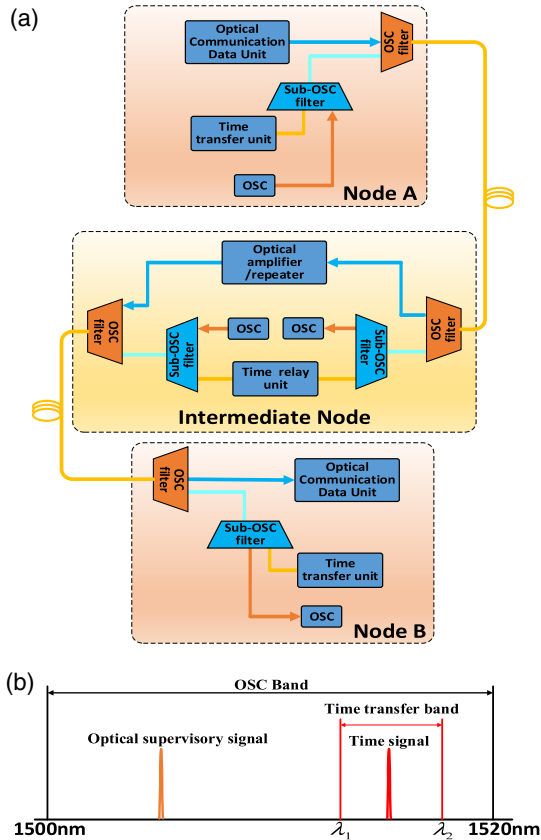


Fig. 1. (a) Diagram of the proposed time transfer through OSC; (b) wavelength arrangement for time signal and optical supervisory signal in the OSC band.

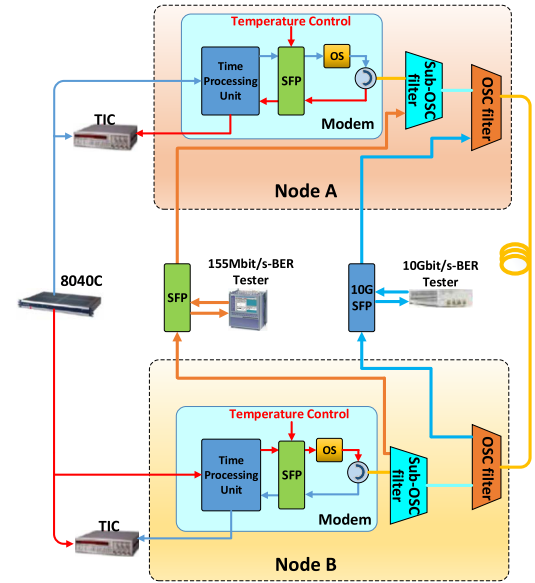


Fig. 2. Experimental setup of time transfer through OSC in commercial WDM systems. OS, optical switch.

The 100 km experimental setup for the proposed scheme is illustrated in Fig. 2. In the experimental system, a bidirectional time division multiplexing transmission over a single fiber with the same wavelength (BTDM-SFSW)-based time transfer scheme [15] is adopted, which only needs one wavelength for the two-way time transfer. Time transfer signals are transmitted through two commercial OSC-band SFPs for conventional optical supervisory signal transmission at two nodes. The two OSC-band SFP cases are tightly attached to external temperature control devices to adjust and stabilize their wavelengths within 1515 ± 0.03 nm. In order to avoid the effect of clock drifts of two different clocks, the one pulse per second (PPS) signal from a common Rb clock (Symmetricom, 8040C) is provided to the time transfer modems at node A and node B simultaneously. Commercial SFPs for optical supervisory signal transmission and 10 Gbit/s dense WDM (DWDM) SFPs with the wavelength of 1550.12 nm are employed for the optical supervisory signal transmission and communication data transmission as usual, respectively. A custom-made sub-OSC filter with a passband range of 1513.31–1521.23 nm is used for multiplexing/demultiplexing the time signal and optical supervisory signal. The commercial OSC filter has a 3 dB passband range of 1501.11–1519.31 nm. Time interval counters (TIC, SR620) are used to measure the time intervals. The BERs of the optical supervisory signal and communication data transmission are tested by a 155 Mbit/s BER tester (Advanced Network Tester-20) and a 10 Gbit/s BER tester (Agilent N4960A/N4951B/N4952A), respectively. The experimental setup is in an ordinary air-conditioned room within temperature fluctuation of around 3°C.

In the BTDM-SFSW-based time transfer scheme, the time difference between two nodes can be expressed as [6]

$$\Delta T = \frac{1}{2}[(T_{AB} - T_{BA} - T_d) + (\tau_{AB}^F - \tau_{BA}^F) + (\tau_A^T - \tau_A^R + \tau_B^R - \tau_B^T)], \quad (1)$$

where $T_{AB}(T_{BA})$ is the time difference between local input 1 PPS and received 1 PPS at node A (node B), T_d is the time delay of local time signal at node B, $\tau_{AB}^F(\tau_{BA}^F)$ is the propagation delay of the fiber link from node A (node B) to node B (node A), and $(\tau_{AB}^F - \tau_{BA}^F)$ can be considered zero in slowly varying circumstances [6]. $\tau_A^T(\tau_B^T)$ and $\tau_A^R(\tau_B^R)$ are the sending and receiving delays at node A (node B), respectively, and can be calibrated by high-precision measurements.

Figure 3(a) shows the measured wavelengths of three employed commercial OSC-band SFPs at different temperatures. Within the operation temperature range of the three SFPs, 0–70°C, the wavelengths of them almost linearly change between 1511.23–1517.21 nm, 1510.68–1517.34 nm, and 1506.62–1513.01 nm, respectively, with the approximately same slope of about 0.09 nm/°C. In the experiment, SFP A and SFP B are used for forward and backward time transmission, and their wavelengths are adjusted and stabilized at 1515 ± 0.03 nm by temperature control. In this case, the wavelength of SFP C in 0–70°C will not overlap that of SFP A and SFP B, and SFP C can be used for optical supervisory signals transmission as in commercial WDM systems. As show in Fig. 3(b), for 24 h in a normal air-conditioned room, both the wavelength fluctuation and wavelength difference of SFP A and SFP B with temperature control are always lower than ± 0.03 nm. If the dispersion coefficient is 17 ps/nm/km, the 100 km two-way delay asymmetry caused by dispersion is less than 102 ps, and the caused uncertainty is less than 51 ps according to Eq. (1) in Ref. [6]. OSC-band SFPs with internal

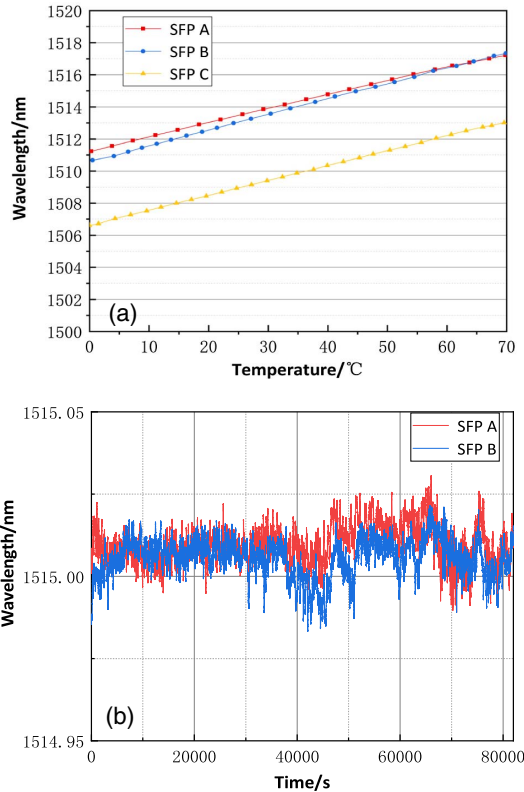


Fig. 3. (a) Measured wavelengths of three commercial OSC-band SFPs as a function of temperatures; (b) the measured wavelength variations of two commercial OSC-band SFPs employed for time transfer during 24 h.

temperature control similar to DWDM SFPs can be adopted to reach the higher wavelength precision and stability for higher-precision time transfer.

The spectrum of the employed commercial OSC filter and the customized sub-OSC filter are shown in Fig. 4. The 3 dB passband of the OSC filter, 1501.11–1519.31 nm, is used for the OSC band, while its reflection band outside the passband is used for data communication, as in commercial WDM systems. The customized sub-OSC filter has a 3 dB passband of 1513.19–1520.99 nm and a reflection band outside the passband. By using sub-OSC filters, the OSC passband of the OSC filter, 1501.11–1519.31 nm, is split into two sub-passbands of 1501.11–1513.02 nm and 1513.19–1519.31 nm for carrying optical supervisory signals and time signals, respectively.

Figure 5(a) shows the measured clock differences obtained by 100 km time transfer. The sending and receiving delays of two nodes are calibrated [1,6]. The theoretical clock differences should be zero for using a common clock at two nodes. We can

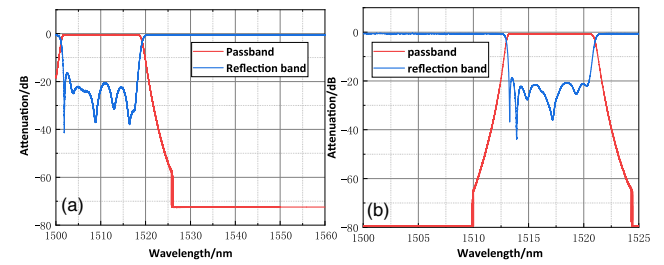


Fig. 4. Spectrum of (a) the employed commercial OSC filter and (b) the customized sub-OSC filter.

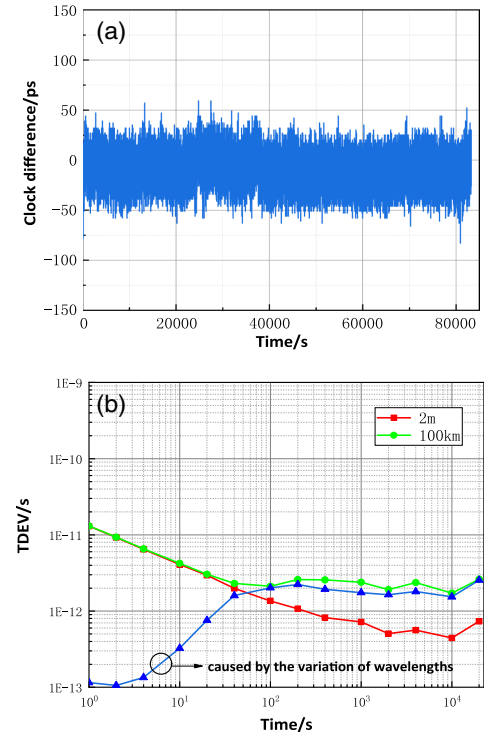


Fig. 5. (a) Measured clock differences of 100 km time transfer; (b) time stability (TDEV) of 100 km time transfer.

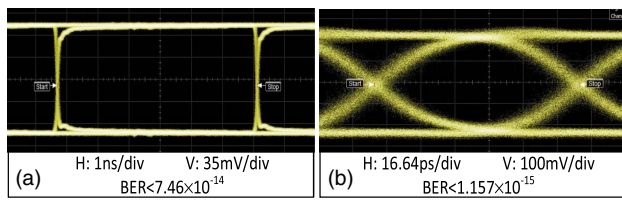


Fig. 6. (a) Eye diagram of optical supervisory channel; (b) eye diagram of communication data channel.

see that the obtained clock differences vary around zero with a peak-to-peak value of less than 142 ps in one day measurement. The variations of obtained clock difference are related to the slight difference of forward and backward wavelength, the delay change of time modem with environment, and so on. The time transfer stabilities (TDEVs) [16] of the 2 m and 100 km fiber link are shown in Fig. 5(b). The TDEVs over the 2 m fiber link are better than 15 ps/s and 0.5 ps/10⁴ s, which can be considered as the noise floor of the system due to very short fiber. The noise floor is mainly determined by the codec scheme, the SFP modules, and the measuring instruments used by the two nodes. The TDEVs over the 100 km fiber link are better than 15 ps/s and 2 ps/10⁴ s. There is almost no change in the short-term stability of time transfer. The result indicates that the influence of the signal-to-noise ratio (SNR) on the time signal is still less than that of the noise floor of the system in 100 km, considering the relationship between the receiving SNR and short-term stability [17]. The deterioration of the long-term stability is mainly caused by the variation of wavelengths of the SFPs with the temperature. From Eq. (1), bidirectional time differences caused by wavelength differences can be expressed as

$$x = \frac{1}{2} \times D \times L \times \Delta\lambda, \quad (2)$$

where D is the dispersion coefficient, L is the optical fiber length, and $\Delta\lambda$ is the forward and backward wavelength difference. The blue curve in Fig. 5(b) shows the TDEV caused by the variation of wavelengths of the SFPs in our experiment when D is 17 ps/nm/km and L is 100 km.

Figure 6 shows the eye diagrams of the 155 Mbit/s optical supervisory signal and the 10 Gbit/s data signal after 100 km transmission, respectively, with a $2^7 - 1$ pseudorandom binary sequence. It can be seen that both eye diagrams after 100 km transmission are clearly identifiable. For either 100 km optical supervisory signal transmission or communication data transmission, there is no bit error being observed in more than 24 h. That means BERs of 100 km optical supervisory signal transmission and communication data transmission can be less than 7.46×10^{-14} and 1.157×10^{-15} , respectively, which meet the requirements of WDM system communication.

In summary, we present a time transfer scheme through OSC in commercial WDM systems and set up a 100 km

experimental system based on the proposed scheme in the laboratory. A sub-band in the OSC band is split out by only adding sub-OSC filters for time transfer, while the optical supervisory signal and existing communication data transmission are not affected. The wavelengths for two-directional transmission can be very close, or even the same, to realize high-precision time transfer without requiring link calibrations. A 100 km combined time, optical supervisory signal, and communication data transmission over a WDM system is experimentally demonstrated. The peak-to-peak value of the measured clock difference and the time deviations of time transfer can be less than 142 ps, 15 ps/s, and 2 ps/10⁴ s, respectively. The BERs of both the OSC and the communication data channel meet communication standards.

Funding. National Natural Science Foundation of China (61535006, 61627817).

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